Supplementary Material

Self-Oscillating Joule Thief Converters for MEMS Electromagnetic Energy Harvesting: Analytical Modeling and Experimental Validation

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1 SPICE Simulation Netlists

This section provides the SPICE netlists used for circuit simulations in the main manuscript. These netlists enable reproduction of the simulation results presented in Sections II and IV.

1.1 BJT-based Converter (LTspice)

The following netlist implements the BJT-based Joule Thief converter in LTspice. The coupled inductors are modeled using the `K` statement with coupling coefficient $k = 0.99$. The simulation models for the SMMBT3904LT1G and RB521S30T1G can be obtained from the manufacturer's official website.

```
V1 1 0 {Voc}
L2 2 1 {a*a*Lc} Rser=2.35
L1 1 4 {Lc} Rser=2.35
C2 out 0 10u
R1 2 3 {Rb}
D1 4 out rb521s30t1
R2 out 0 {RL}
Q1 4 3 0 0 mmbt3904lt1g
K12 L1 L2 0.99
.tran 0 1 0.9 1e-7 startup
.param RL=1k Rb=10k Lc=1m a=1 Voc=1
.meas TRAN V_out AVG V(out)
```

Listing 1: LTspice netlist for BJT-based converter

Parameter definitions:

- `Voc`: Open-circuit source voltage
- `Lc`: Primary inductance L_1
- `a`: Turns ratio ($L_2 = a^2 L_1$)
- `Rb`: Base resistance
- `RL`: Load resistance

1.2 NMOS-based Converter (Cadence Spectre)

The following netlist implements the NMOS-based Joule Thief converter in Cadence Spectre, using GlobalFoundries' 180 nm process with native NMOS devices.

```
simulator lang=spectre
global 0 vdd!
parameters Wn=10u Ln=1u RL=10K Rin=200
parameters k0=0.99 a=1 L0=1m Vin=200m
parameters CL=10u Md=50 nf=50
NM1 (drain1 gate1 0 0) nmos_1p8_nat_iso
    w=(Wn)*(nf) l=Ln m=1 nf=nf
R1 (out 0) resistor r=RL
R0 (vdd! V_in) resistor r=Rin
K1 mutual_inductor coupling=k0
    ind1=L1 ind2=L2
L2 (gate1 vdd!) inductor l=a*a*L0 r=2.35
L1 (vdd! drain1) inductor l=L0 r=2.35
V0 (V_in 0) vsource dc=Vin type=dc
D0 (drain1 out) sc_diode
    area=12.4p pj=41.24u m=Md
C0 (out 0) capacitor c=CL
ic gate1=0
tran tran stop=1 start=0 outputstart=0.9
```

Listing 2: Spectre netlist for NMOS-based converter

Parameter definitions:

- `Wn`, `Ln`: NMOS channel width and length
- `nf`: Number of fingers
- `Rin`: Source impedance
- `Vin`: Input voltage (equivalent to V_{oc})
- `k0`: Coupling coefficient
- `a`: Turns ratio
- `L0`: Primary inductance L_1
- `RL`: Load resistance
- `CL`: Output capacitance
- `Md`: Diode multiplier

2 BJT Peak Current Derivation

This section derives the BJT peak current condition presented in Section II of the main text.

During BJT conduction, $i_1 = i_{\mathrm{CE}}$ and $i_2 = i_{\mathrm{BE}}$. Applying Kirchhoff's voltage law (KVL) yields:

$$V_{\mathrm{oc}} - V_{\mathrm{L1}}(t) = V_{\mathrm{CE}}(t) \tag{S.1}$$

$$V_{\mathrm{oc}} - V_{\mathrm{L2}}(t) - R_{\mathrm{b}} i_{\mathrm{BE}}(t) = V_{\mathrm{BE}}(t) \tag{S.2}$$

Since base current dynamics are slower than collector current dynamics, and $i_2 = i_{\mathrm{BE}} \approx i_{\mathrm{CE}}/\beta_{\mathrm{F}}$ with $\beta_{\mathrm{F}} \gg 1$, we have:

$$V_{\mathrm{L1}}(t) \approx -\frac{1}{a} V_{\mathrm{L2}}(t) \tag{S.3}$$

Substituting into Eqs. (S.1)–(S.2), we derive:

$$V_{\mathrm{BE}} + aV_{\mathrm{CE}} = (1+a)V_{\mathrm{oc}} - R_{\mathrm{b}} i_{\mathrm{BE}} \tag{S.4}$$

Using the Ebers-Moll model, the base and collector currents are:

$$i_{\mathrm{BE}} = \frac{I_{\mathrm{s}}}{\beta_{\mathrm{F}}}\left[e^{V_{\mathrm{BE}}/V_{\mathrm{T}}} - 1\right] + \frac{I_{\mathrm{s}}}{\beta_{\mathrm{R}}}\left[e^{V_{\mathrm{BC}}/V_{\mathrm{T}}} - 1\right] \tag{S.5}$$

$$i_{\mathrm{CE}} = I_{\mathrm{s}}\left[e^{V_{\mathrm{BE}}/V_{\mathrm{T}}} - e^{V_{\mathrm{BC}}/V_{\mathrm{T}}}\right] - \frac{I_{\mathrm{s}}}{\beta_{\mathrm{R}}}\left[e^{V_{\mathrm{BC}}/V_{\mathrm{T}}} - 1\right] \tag{S.6}$$

where V_{T} is thermal voltage, $\beta_{\mathrm{F}}/\beta_{\mathrm{R}}$ are forward/reverse current gains, I_{s} is saturation current, and $V_{\mathrm{BC}} = V_{\mathrm{BE}} - V_{\mathrm{CE}}$.

Substituting Eqs. (S.5)–(S.6) into Eq. (S.4):

$$\begin{aligned} V_{\mathrm{BE}} + aV_{\mathrm{CE}} = (1+a)V_{\mathrm{oc}} &- \frac{R_{\mathrm{b}} I_{\mathrm{s}}}{\beta_{\mathrm{F}}}\left[e^{V_{\mathrm{BE}}/V_{\mathrm{T}}} - 1\right] \\ &- \frac{R_{\mathrm{b}} I_{\mathrm{s}}}{\beta_{\mathrm{R}}}\left[e^{(V_{\mathrm{BE}} - V_{\mathrm{CE}})/V_{\mathrm{T}}} - 1\right] \end{aligned} \tag{S.7}$$

During the turn-on phase, the rising collector current induces a positive primary voltage $V_{\mathrm{L1}} \approx L_1 \cdot \mathrm{d}i_{\mathrm{CE}}/\mathrm{d}t$. Through the inverse magnetic coupling [Eq. (S.3)], this induces a negative secondary voltage V_{L2}. From Eq. (S.2), the negative V_{L2} drives V_{BE} to rise, establishing positive feedback. As the BJT approaches saturation, the current slope $\mathrm{d}i_{\mathrm{CE}}/\mathrm{d}t$ gradually decreases. At $t = t_{\mathrm{on}}$, the collector current reaches its peak i_{peak} where $\mathrm{d}i_{\mathrm{CE}}/\mathrm{d}t = 0$. At this instant, V_{L1} and V_{L2} both vanish, the feedback ceases, and V_{BE} reaches its peak value simultaneously:

$$\left.\frac{\mathrm{d}V_{\mathrm{BE}}}{\mathrm{d}t}\right|_{t=t_{\mathrm{on}}} = 0 \tag{S.8}$$

Since both V_{BE} and V_{CE} are time functions and V_{CE} is still increasing at this instant ($\mathrm{d}V_{\mathrm{CE}}/\mathrm{d}t \neq 0$), the condition (S.8) implies:

$$\left.\frac{\partial V_{\mathrm{BE}}}{\partial V_{\mathrm{CE}}}\right|_{t=t_{\mathrm{on}}} = \frac{\mathrm{d}V_{\mathrm{BE}}/\mathrm{d}t}{\mathrm{d}V_{\mathrm{CE}}/\mathrm{d}t} = 0 \tag{S.9}$$

Considering V_{BE} as an implicit function of V_{CE} in Eq. (S.7) and applying Eq. (S.9), we obtain the base-emitter voltage at peak current:

$$V_{\mathrm{BE}}^* = V_{\mathrm{CE}}^* + V_{\mathrm{T}} \ln\left(\frac{a\beta_{\mathrm{R}} V_{\mathrm{T}}}{I_{\mathrm{s}} R_{\mathrm{b}}}\right) \tag{S.10}$$

where the superscript $*$ denotes quantities evaluated at the peak current condition. Substituting into Eq. (S.7) yields a transcendental equation solvable using the Lambert W function[1] $W(x)$, defined as the inverse of $f(w) = we^w$, i.e., $W(x)e^{W(x)} = x$. For $x \geq -1/e$, the principal branch $W_0(x)$ provides a single real-valued solution:

$$V_{\mathrm{CE}}^* = \xi_1 - V_{\mathrm{T}} \cdot W(\xi_2) \tag{S.11}$$

[1] T. C. Banwell and A. Jayakumar, "Exact analytical solution for current flow through diode with series resistance," *Electron. Lett.*, vol. 36, no. 4, pp. 291–292, Feb. 2000.

where

$$\xi_1 = V_{\mathrm{oc}} + \frac{I_{\mathrm{s}} R_{\mathrm{b}}}{1+a}\left(\frac{1}{\beta_{\mathrm{F}}} + \frac{1}{\beta_{\mathrm{R}}}\right) - \frac{V_{\mathrm{T}}}{1+a}\left[a + \ln\left(\frac{a V_{\mathrm{T}} \beta_{\mathrm{R}}}{I_{\mathrm{s}} R_{\mathrm{b}}}\right)\right]$$

$$\xi_2 = \frac{a\beta_{\mathrm{R}}}{(1+a)\beta_{\mathrm{F}}} \exp\left(\frac{\xi_1}{V_{\mathrm{T}}}\right)$$

With V^*_{CE} and V^*_{BE} determined, the peak current i_{peak} follows from Eqs. (S.5)–(S.6) by substituting $V^*_{\mathrm{BC}} = V^*_{\mathrm{BE}} - V^*_{\mathrm{CE}}$:

$$i_{\mathrm{peak}} = I_{\mathrm{s}}\left[e^{V^*_{\mathrm{BE}}/V_{\mathrm{T}}} - e^{V^*_{\mathrm{BC}}/V_{\mathrm{T}}}\right] - \frac{I_{\mathrm{s}}}{\beta_{\mathrm{R}}}\left[e^{V^*_{\mathrm{BC}}/V_{\mathrm{T}}} - 1\right] \tag{S.12}$$

3 NMOS Peak Current Derivation

This section derives the NMOS peak current condition presented in Section II of the main text.

Applying Kirchhoff's voltage law (KVL) to the drain and gate loops of the NMOS-based circuit, we have:

$$V_{\mathrm{oc}} - R_{\mathrm{in}}(i_1 + i_2) - V_{\mathrm{L1}} = V_{\mathrm{ds}} \tag{S.13}$$

$$V_{\mathrm{oc}} - R_{\mathrm{in}}(i_1 + i_2) - V_{\mathrm{L2}} = V_{\mathrm{gs}} \tag{S.14}$$

During conduction, the gate current is negligible ($i_2 \approx 0$), thus $i_1 = i_{\mathrm{ds}}$. Using the coupled inductor relationship $V_{\mathrm{L2}} \approx -aV_{\mathrm{L1}}$ and eliminating V_{L1} from Eqs. (S.13)–(S.14), we derive the voltage relationship:

$$(1 + a)V_{\mathrm{oc}} - aV_{\mathrm{ds}} - (1 + a)R_{\mathrm{in}}i_{\mathrm{ds}} = V_{\mathrm{gs}} \tag{S.15}$$

With the inductor current increasing ($V_{\mathrm{L1}} > 0$), the NMOS operates in the triode region since $V_{\mathrm{gs}} - V_{\mathrm{ds}} \approx (1 + a)V_{\mathrm{L1}}$ exceeds the threshold voltage V_{th}. The drain-source current in the triode region is:

$$i_{\mathrm{ds}} = \beta_{\mathrm{n}} V_{\mathrm{ds}} \left[V_{\mathrm{gs}} - V_{\mathrm{th}} - V_{\mathrm{ds}}/2\right] \tag{S.16}$$

where $\beta_{\mathrm{n}} = \mu_{\mathrm{n}} C_{\mathrm{ox}}(W_{\mathrm{n}}/L_{\mathrm{n}})$ is the transconductance parameter.

Substituting Eq. (S.15) into Eq. (S.16) and performing algebraic manipulations:

$$i_{\mathrm{ds}} = \frac{\beta_{\mathrm{n}} V_{\mathrm{ds}} \left[2(a + 1)V_{\mathrm{oc}} - (2a + 1)V_{\mathrm{ds}} - 2V_{\mathrm{th}}\right]}{2[1 + (a + 1)\beta_{\mathrm{n}} V_{\mathrm{ds}} R_{\mathrm{in}}]} \tag{S.17}$$

At the peak current, $\partial i_{\mathrm{ds}}/\partial t = 0$ while $\partial V_{\mathrm{ds}}/\partial t \neq 0$, which implies $\partial i_{\mathrm{ds}}/\partial V_{\mathrm{ds}} = 0$. Taking the partial derivative of Eq. (S.17) with respect to V_{ds} and setting it to zero yields:

$$V_{\mathrm{ds}}^{*} = \frac{\sqrt{b_1(2\beta_{\mathrm{n}} b_2 b_3 + b_1)} - b_1}{\beta_{\mathrm{n}} b_1 b_2} \tag{S.18}$$

$$i_{\mathrm{peak}} = \frac{\beta_{\mathrm{n}} b_2 b_3 + b_1 - \sqrt{b_1(2\beta_{\mathrm{n}} b_2 b_3 + b_1)}}{\beta_{\mathrm{n}} b_2^2} \tag{S.19}$$

where $b_1 = 1 + 2a$, $b_2 = (a + 1)R_{\mathrm{in}}$, $b_3 = (a + 1)V_{\mathrm{oc}} - V_{\mathrm{th}}$.

For an ideal source ($R_{\mathrm{in}} = 0$), these simplify to:

$$V_{\mathrm{ds}}^{*} = \frac{(a + 1)V_{\mathrm{oc}} - V_{\mathrm{th}}}{2a + 1}, \quad i_{\mathrm{peak}} = \frac{\beta_{\mathrm{n}}[(a + 1)V_{\mathrm{oc}} - V_{\mathrm{th}}]^2}{2(2a + 1)}$$

4 MATLAB Calculation Scripts

This section provides the MATLAB functions used to verify the analytical model. The variable names correspond to the notation in Table II.

4.1 Script 1: BJT-Based Converter

```
function [V_out, freq, P_in, P_out, eff] = calculate_bjt_full(V_oc, a, L_1, R_L, R_b, I_s, beta_F, beta_R, V_T, V_D)
    % Calculates steady-state performance of BJT Joule Thief (Section II-B).
    %
    % Inputs:
    %   V_oc   : Open-circuit source voltage [V]
    %   a      : Transformer turns ratio (N2/N1)
    %   L_1    : Primary inductance [H]
    %   R_L    : Load resistance [Ohm]
    %   R_b    : Base resistance [Ohm]
    %   I_s    : Saturation current [A]
    %   beta_F : Forward current gain
    %   beta_R : Reverse current gain
    %   V_T    : Thermal voltage [V] (~26mV at 300K)
    %   V_D    : Diode forward voltage drop [V]
    %
    % Outputs:
    %   V_out  : Output voltage across R_L [V]
    %   freq   : Switching frequency [Hz]
    %   P_in   : Input power [W]
    %   P_out  : Output power [W]
    %   eff    : Power conversion efficiency

    % 1. Calculate Peak Voltage Condition
    % xi_1 and xi_2 per Eq. (S.11) in this supplementary material
    xi_1 = V_oc + (I_s .* R_b ./ (1 + a)) .* (1 ./ beta_F + 1 ./ beta_R) - ...
           (V_T ./ (1 + a)) .* (a + log((a .* V_T .* beta_R) ./ (I_s .* R_b)));

    xi_2 = (a .* beta_R) ./ ((1 + a) .* beta_F) .* exp(xi_1 ./ V_T);

    % Solve for V_CE* at peak current per Eq. (S.11)
    V_CE = xi_1 - V_T .* lambertw(xi_2);

    % 2. Calculate V_BE* per Eq. (S.10)
    V_BE = log((a .* V_T .* beta_R) ./ (I_s .* R_b)) .* V_T + V_CE;

    % 3. Calculate peak currents using Ebers-Moll model
    % i_BE per Eq. (S.5), i_peak per Eq. (S.12)
    i_BE = I_s .* ((-1 + exp(V_BE ./ V_T)) ./ beta_F + ...
           (-1 + exp((V_BE - V_CE) ./ V_T)) ./ beta_R);
    i_peak = I_s .* (exp(V_BE ./ V_T) - exp((V_BE - V_CE) ./ V_T) - ...
             (-1 + exp((V_BE - V_CE) ./ V_T)) ./ beta_R);

    % 4. Calculate Turn-on Time (Eq. 8 in main text)
    t_on = (2 .* i_peak .* L_1) ./ (2 .* V_oc - V_CE);

    % 5. Calculate Turn-off Time and Output Voltage (Eqs. 12-13 in main text)
    gamma_1 = (2 .* (V_oc - V_D)) ./ (2 .* V_oc - V_CE);
    gamma_2 = ((2 .* V_D - V_CE).^2 + 4 .* i_peak .* R_L .* (2 .* V_oc - V_CE)) ./ ...
              (2 .* V_oc - V_CE).^2;

    V_out = (V_oc - V_D) + (i_peak .* R_L - 2 .* (V_oc - V_D)) ./ ...
            (1 + gamma_1 + sqrt(gamma_2));

    t_off = (i_peak .* L_1 .* (1 + gamma_1 + sqrt(gamma_2))) ./ ...
```

```
            (i_peak .* R_L + 2 .* (V_D - V_oc));

    % 6. Performance Metrics
    freq = 1 ./ (t_on + t_off);      % Switching Frequency

    % Input Power (Triangular Waveform Approx)
    P_in = V_oc .* (i_peak + i_BE) ./ 2;

    P_out = V_out.^2 ./ R_L;         % Output Power
    eff = P_out ./ P_in;             % Efficiency
end
```

4.2 Script 2: NMOS-Based Converter ($R_{\text{in}} > 0$)

```
function [V_out, freq, P_in, P_out, eff] = calculate_nmos_resistive(V_oc, R_in,
    a, L_1, R_L, beta_n, V_th, V_D)
    % Calculates steady-state performance of NMOS Joule Thief (Section II-C).
    % Covers the general case with non-zero source resistance (Rin > 0).
    %
    % Inputs:
    %   V_oc    : Open-circuit voltage of the source [V]
    %   R_in    : Source internal resistance [Ohm]
    %   a       : Transformer turns ratio (N2/N1)
    %   L_1     : Primary inductance [H]
    %   R_L     : Load resistance [Ohm]
    %   beta_n  : NMOS transconductance parameter (mu_n * Cox * W/L) [A/V^2]
    %   V_th    : NMOS Threshold voltage [V]
    %   V_D     : Diode forward voltage drop [V]
    %
    % Outputs:
    %   V_out   : Output voltage across R_L [V]
    %   freq    : Switching frequency [Hz]
    %   P_in    : Input power [W]
    %   P_out   : Output power [W]
    %   eff     : Power conversion efficiency

    % 1. Calculate Unified Model Coefficients
    % b1, b2, b3 per Eqs. (S.18)-(S.19) in this supplementary material
    b1 = 1 + 2 .* a;
    b2 = (a + 1) .* R_in;
    b3 = (a + 1) .* V_oc - V_th;

    % 2. Calculate V_ds* at peak current per Eq. (S.18)
    V_ds_star = (sqrt(b1 .* (2 .* beta_n .* b2 .* b3 + b1)) - b1) ./ ...
                (beta_n .* b1 .* b2);

    % 3. Calculate peak primary current per Eq. (S.19)
    i_peak = (beta_n .* b2 .* b3 + b1 - sqrt(b1 .* (2 .* beta_n .* b2 .* b3 +
        b1))) ./ ...
             (beta_n .* b2.^2);

    % 4. Calculate Turn-on Time (Eq. 16 in main text)
    t_on = (2 .* L_1 .* i_peak) ./ (2 .* V_oc - i_peak .* R_in - V_ds_star);

    % 5. Calculate Output Voltage and Turn-off Time (Eqs. 20-21 in main text)
    gamma_3 = (V_ds_star - 2 .* V_D).^2 + 4 .* R_L .* i_peak .* ...
              (2 .* V_oc - V_ds_star - i_peak .* R_in);

    V_out = 0.25 .* (V_ds_star - 2 .* V_D + sqrt(gamma_3));

    t_off = (4 .* i_peak .* L_1) ./ (V_ds_star + 2 .* i_peak .* R_in + ...
            2 .* V_D - 4 .* V_oc + sqrt(gamma_3));

    % 6. Performance Metrics
    freq = 1 ./ (t_on + t_off);              % Switching Frequency

    % Input Power (Triangular Waveform Approx)
    i_avg = i_peak ./ 2;
    P_in = (V_oc - R_in .* i_avg) .* i_avg;

    P_out = V_out.^2 ./ R_L;                 % Output Power
    eff = P_out ./ P_in;                     % Efficiency
end
```